

TARGET-90

JKBOSE HUMANITIES CLASS- 11TH

VECTOR (MOTION IN 2D)

One Shot

PHYSICS

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Topics to be Covered

A Vector ✓

B Types of Vector ✓

C Algebra of Vector ✓

D Component of Vector ✓



Vector

Maths is the language of Physics ✓

Math is the language of universe ✓

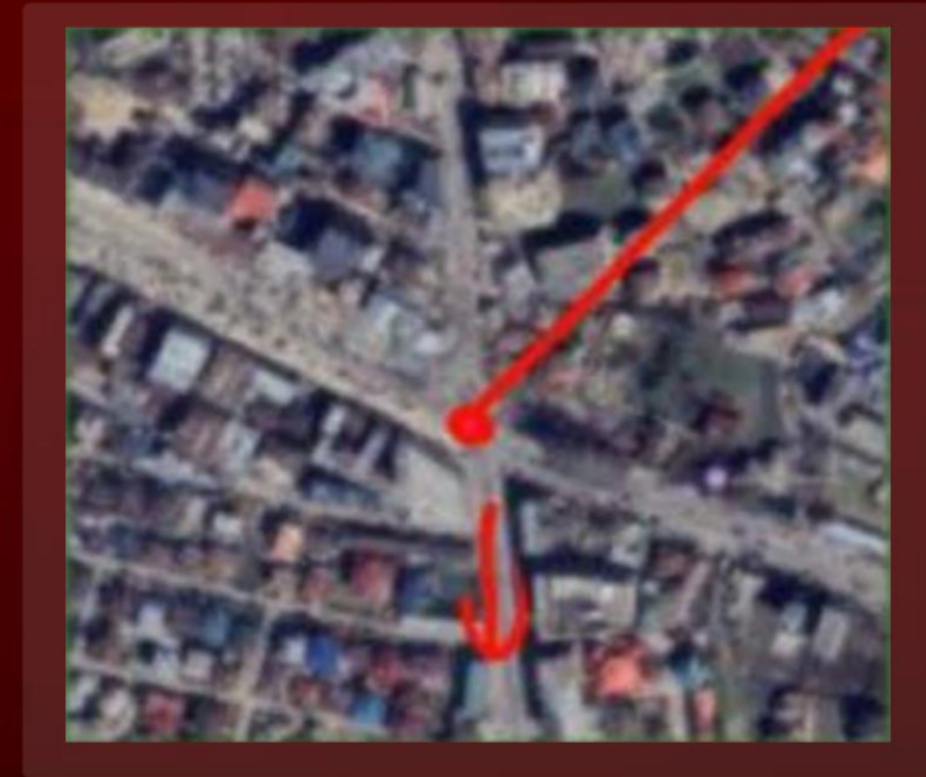
Maths Ke bina Numerical and Derivation nahi honge





Vector

- ◆ The physical quantity which have magnitude & direction
- ◆ **Example:** Displacement $\vec{x}$ velocity, $\vec{v}$
acceleration, Force etc
 $\vec{a}$ $\vec{F}$



Vector = magnitudes × direction

number

Hyderpora	5 km ←
Lal Bazar	4 km ↑
Parray Pora	6 km ↓
Rajbag	7 km →



Feel of vector

Gulmarg = 80km ↓
Airport = 20km →
Pulwama = 10km ↑
✓ Barzulla = 5km ←

← Barzulla

Pulwama

Budgar
Airport

Barzulla = 5km × ←
 ↓
 magnitude Direction

5km



5km

5km

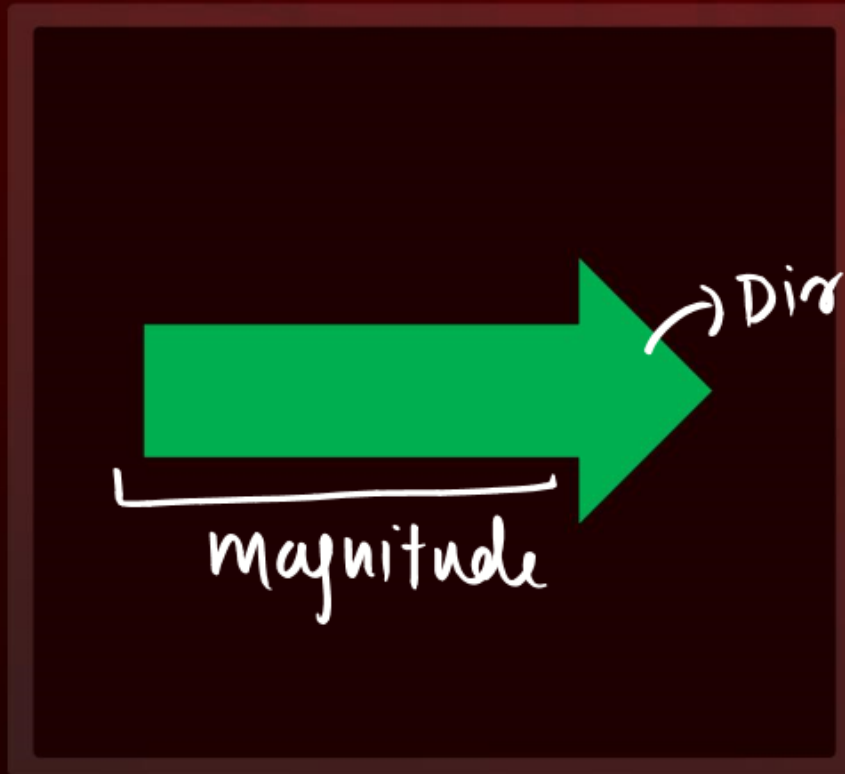
Gulmarg



Motion

→
→
→
→
→

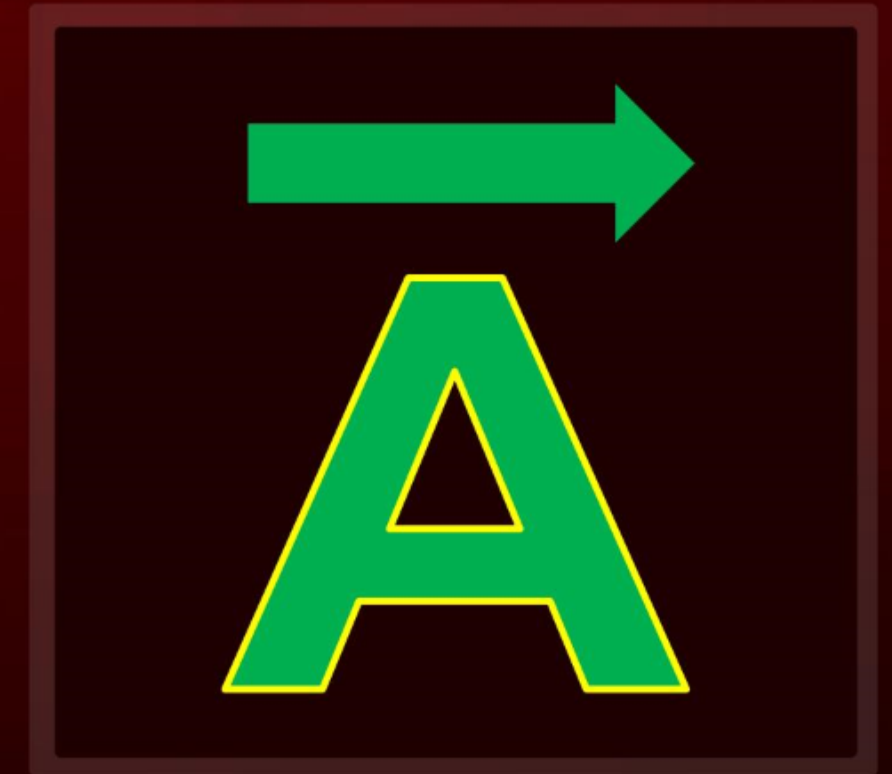
Representation of Vector



By an arrow



Bold Letter



Letter with Arrow

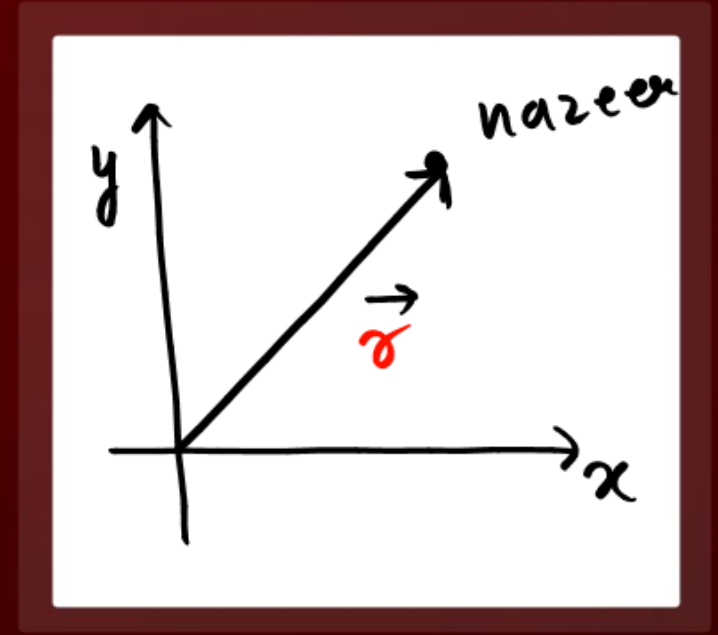


Types of Vector

1. Position Vector



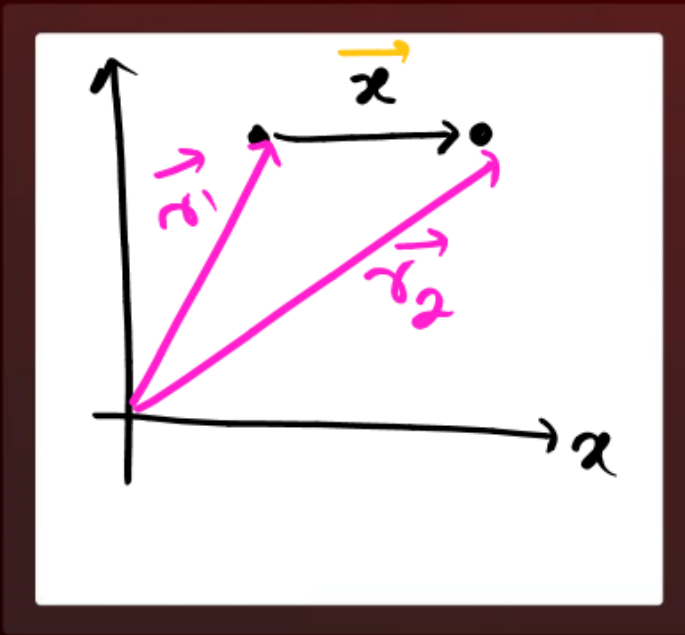
The vector which give the position of the particle



2. Displacement Vector



The vector which shows the displacement.



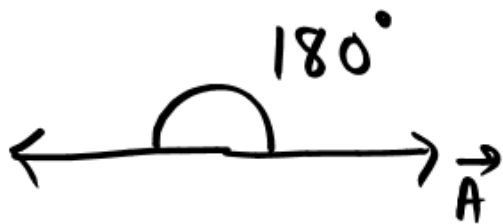


Types of Vector

3. Parallel Vector

Two vector are parallel when the angle between two vector is zero [0°].

$$\begin{array}{c} \longrightarrow \vec{A} \\ \longrightarrow \vec{B} \\ \vec{A} \parallel \vec{B} \\ \theta = 0^\circ \end{array}$$



4. Anti-Parallel Vector

Two vector are Anti-parallel when the angle between two vectors is 180 degrees.



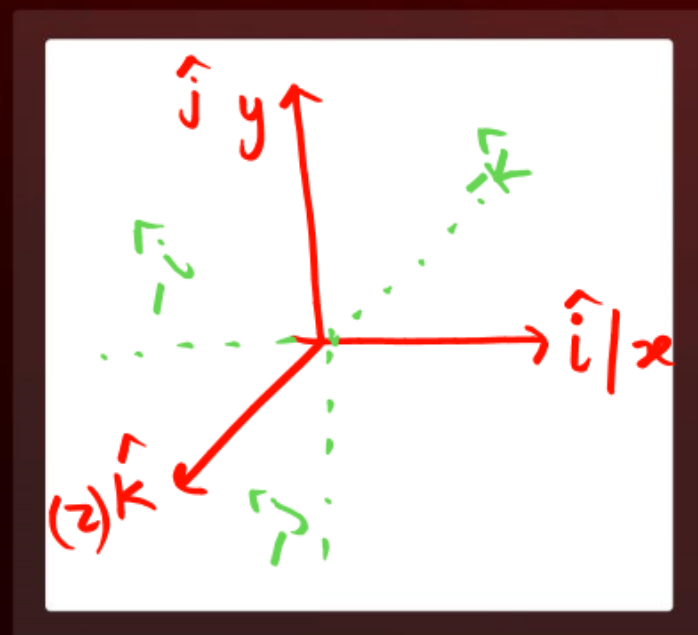
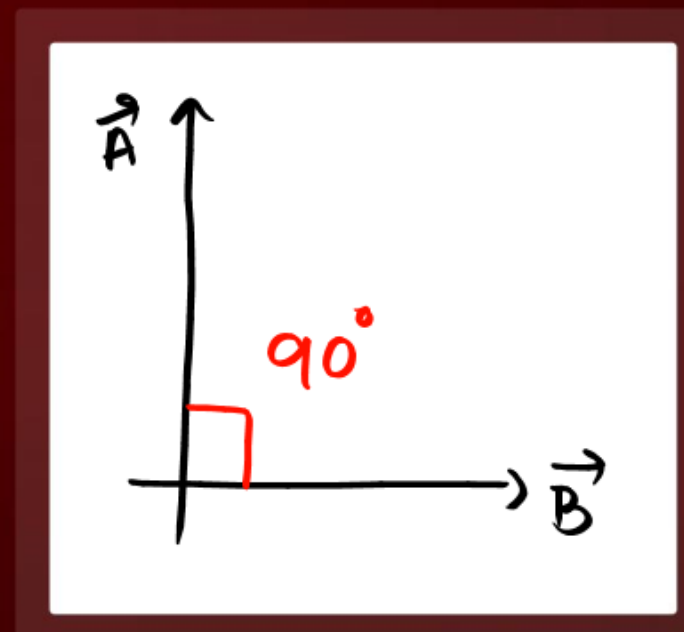
Types of Vector

$$\vec{V} = V \cdot \hat{V}$$

$$\frac{\vec{V}}{V} = \hat{V}$$

5. Orthogonal Vector

Two vector are orthogonal vectors when the angle between two vectors is 90 degrees. / Tarana Rana ✓



6. Unit Vector

The vector which have magnitude is 1 and have only direction.

$$\frac{\vec{A}}{|\vec{A}|} = \frac{A \times \hat{A}}{A} = \hat{A}$$

→ → → → →



Types of Vector

7. Equal Vector

$$\vec{A} = \vec{B}$$

Two vector have same magnitude and direction.

$$\begin{aligned} \longrightarrow \vec{A} &= 10 \rightarrow \\ \longrightarrow \vec{B} &= 10 \rightarrow \\ \vec{A} &= \vec{B} \end{aligned}$$

8. Null/zero Vector

The vector with zero magnitude.

$$\vec{A} = 0\hat{i}$$

$$\vec{A} = 0\hat{A}$$



$$\begin{array}{r} 5421 \\ + 1241 \\ \hline 6661 \end{array}$$

Class - 10th

Scalar

+ Add $2 + 2 = 4$

- Sub $2 - 2 = 0$

x $2 \times 2 = 4$

$\frac{\circ}{\circ}$ $\frac{2}{2} = 1$

Class 11th

Vector

$\vec{A} + \vec{B}$

Add (+)

$\vec{A} - \vec{B}$

(-)

$\vec{A} \times \vec{B}$

$\vec{A} \cdot \vec{B}$

x

$\frac{\vec{A}}{\vec{B}}$

$\div$

Algebra of vector

5 mark

3 mark



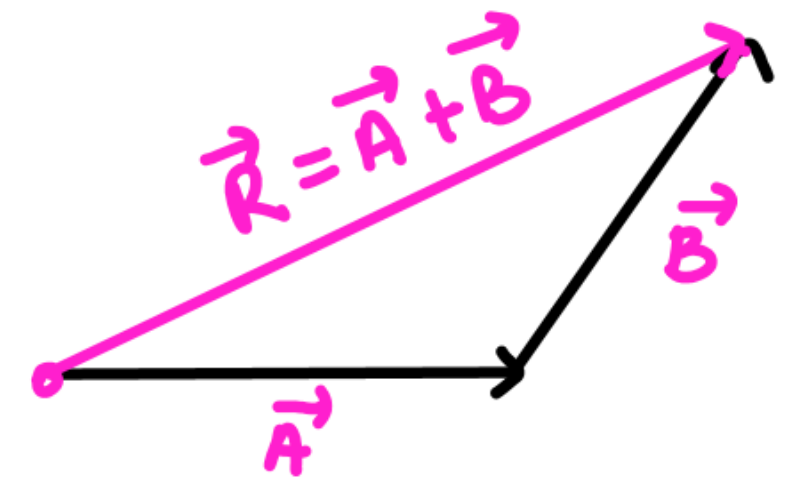
Vector Addition

①

Vector Addition

1. Triangular Law of Vector addition

If Two vectors are the two side of the triangle, then Resultant vector is the 3rd side in opposite Sense.





Vector Addition

Vector Addition

5mark

2. Parallelogm Law of Vector addition

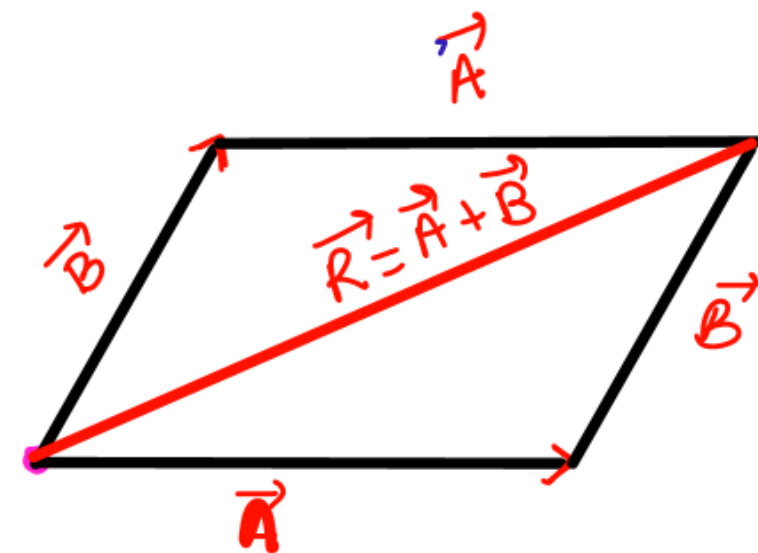
If Two vectors are the adjacent side of parallelogram then Diagonal is the Resultant Vector.

$$\vec{R} = \vec{A} + \vec{B}$$

$$R_{net} = ?$$

$$\tan \alpha = ?$$

2025
2023 2024
2026



(•) Resultant Vector (R)

$$R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

In ΔPQR

$$\sin \theta = \frac{P}{H}$$

$$\cos \theta = \frac{B}{H}$$

$$\sin \theta = \frac{y}{B}$$

$$\cos \theta = \frac{x}{B}$$

$$B \sin \theta = y$$

$$B \cos \theta = x$$

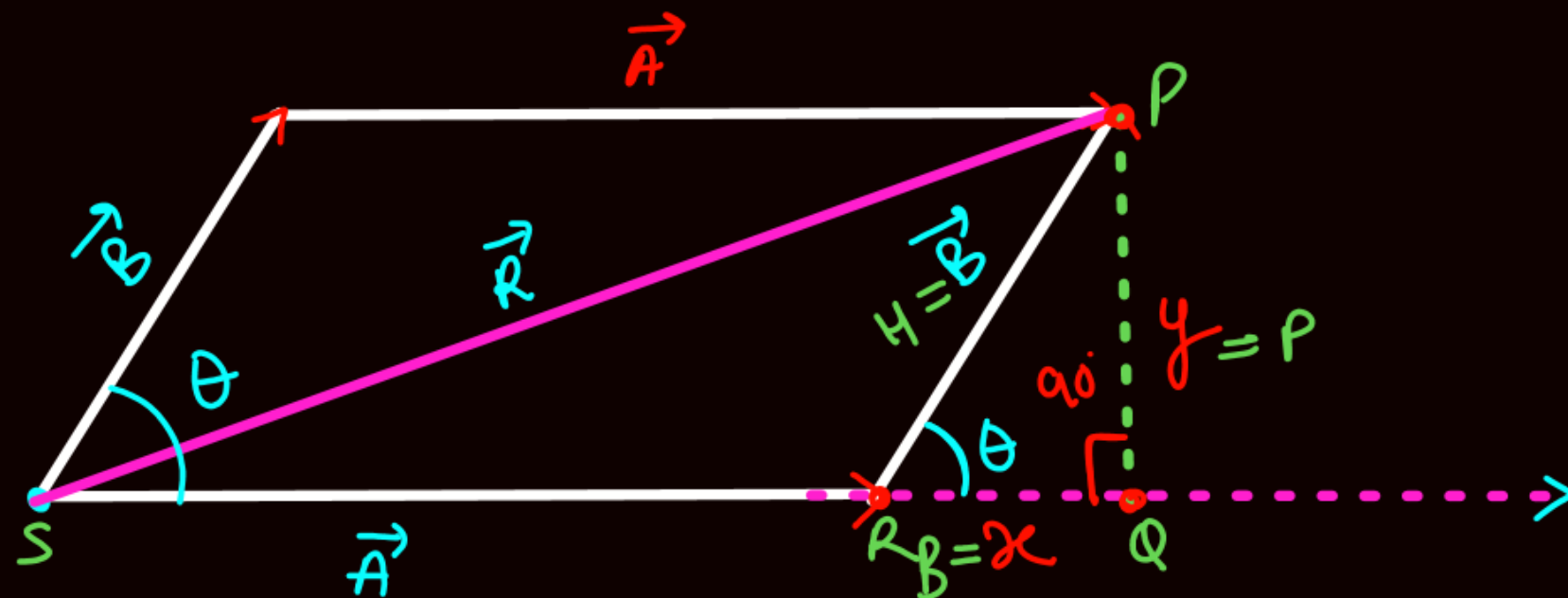
$$x = B \cos \theta$$

In ΔPQS

By PGI

$$H^2 = B^2 + P^2$$

$$R^2 = (A + x)^2 + y^2$$



(•) Resultant Vector (R) $R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

$$R^2 = (A + x)^2 + y^2 \quad (a+b)^2 = a^2 + b^2 + 2ab$$

$$R^2 = (A^2 + x^2 + 2Ax) + y^2$$

$$R^2 = A^2 + (B\cos\theta)^2 + 2AB\cos\theta + (B\sin\theta)^2$$

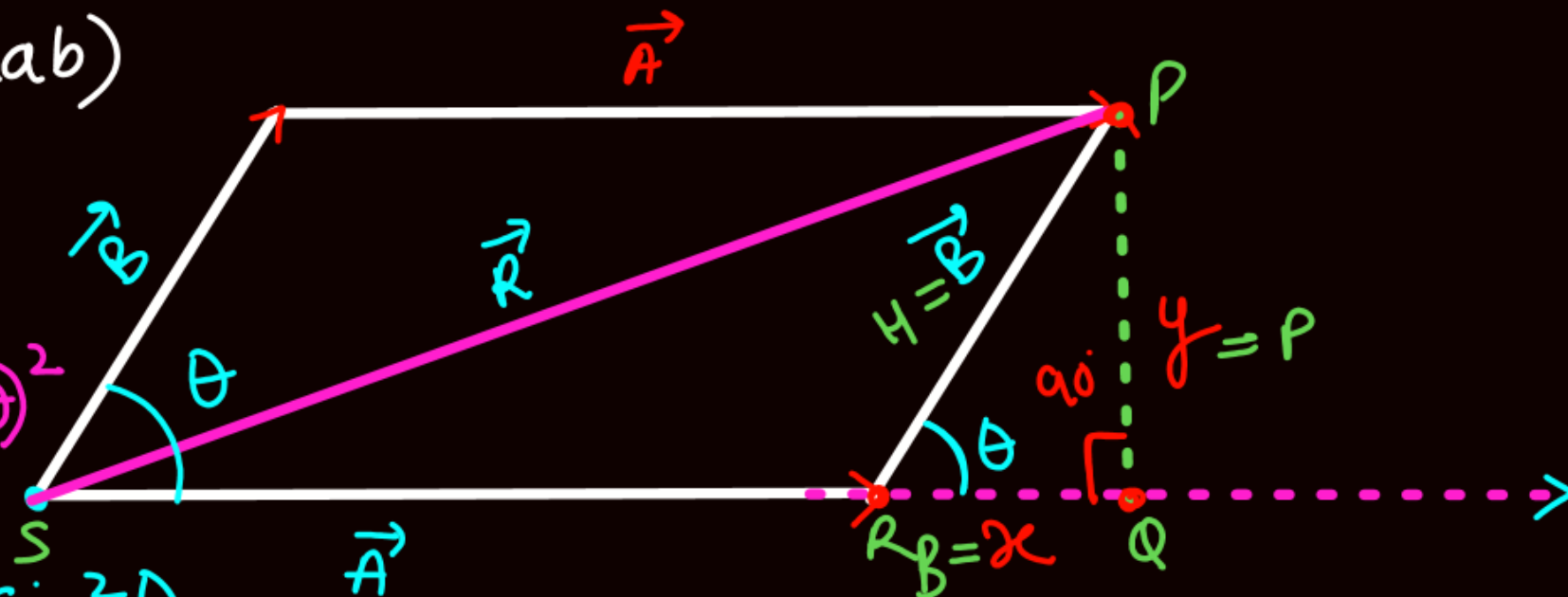
$$R^2 = A^2 + B^2\cos^2\theta + 2AB\cos\theta + B^2\sin^2\theta$$

$$R^2 = A^2 + B^2\sin^2\theta + B^2\cos^2\theta + 2AB\cos\theta$$

$$R^2 = A^2 + B^2(\sin^2\theta + \cos^2\theta) + 2AB\cos\theta$$

$$R^2 = A^2 + B^2 \times 1 + 2AB\cos\theta$$

$$R^2 = A^2 + B^2 + 2AB\cos\theta$$

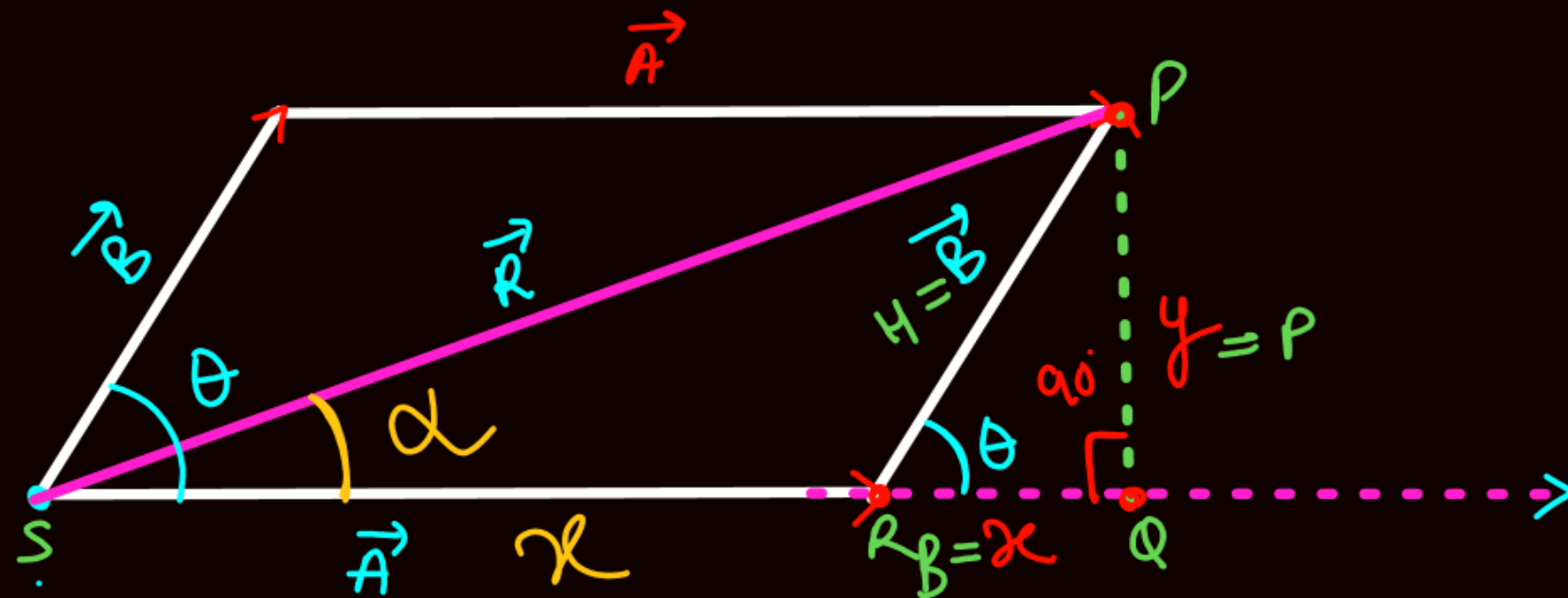


$$\sin^2\theta + \cos^2\theta = 1$$

(•) Resultant Vector (R)

$$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$$

$$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$$



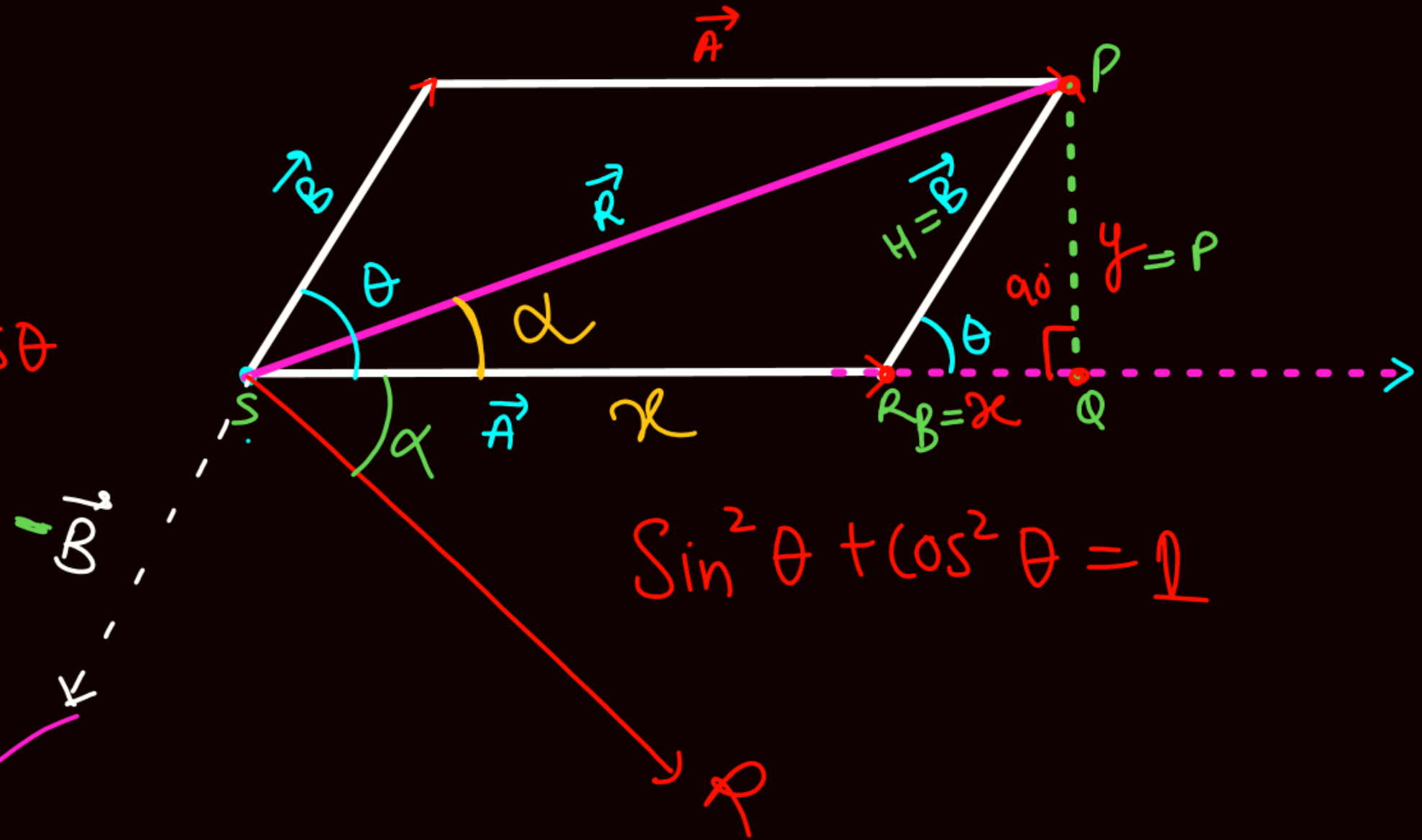
$$\# \tan \alpha = \frac{P}{B}$$

$$\tan \alpha = \frac{y = B \sin \theta}{A + x = B \cos \theta}$$

$$\tan \alpha = \frac{B \sin \theta}{A + B \cos \theta}$$

$$R = \sqrt{A^2 + B^2 - 2AB \cos \theta}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$





(Resolution of Vector)

Component of Vector

2024



Rectangular Component of vector

- Converting the Resultant vector into its component.

$$\sin \theta = \frac{P}{H}$$

$$\sin \theta = \frac{R_y}{R}$$

$$R \sin \theta = R_y$$

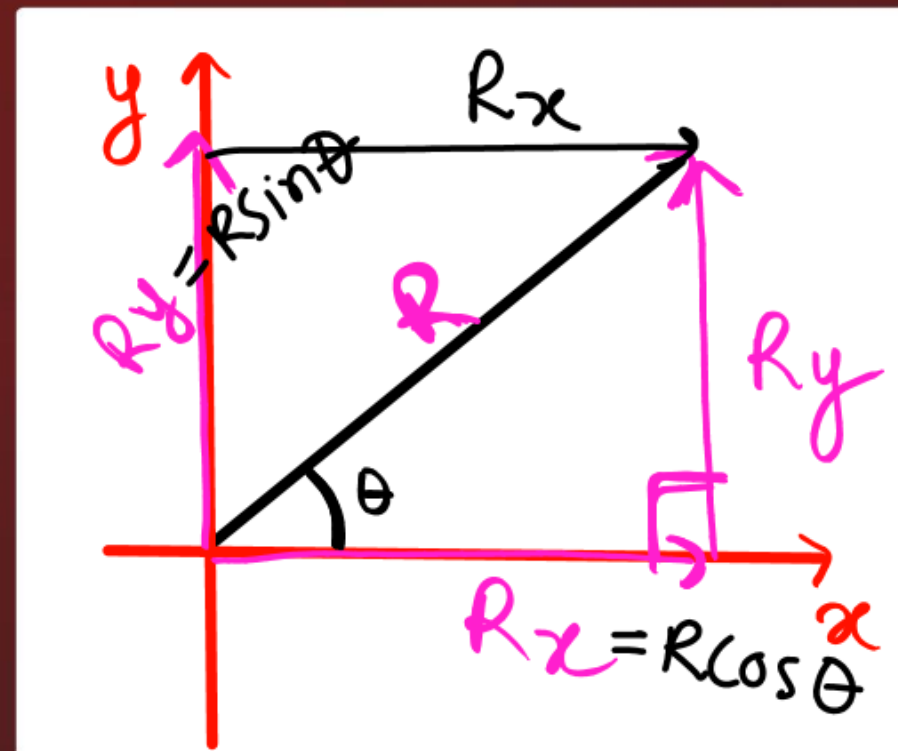
$$\cos \theta = \frac{B}{H}$$

$$\cos \theta = \frac{R_x}{R}$$

$$R \cos \theta = R_x$$

$$R_x = R \cos \theta$$

$R_x = ?$
 $R_y = ?$





Multiplication in vectors

a. Multiplication by constt.

$$\vec{A} \times 2 = 2\vec{A}$$


no change
in direction

$$\vec{A} \times -2 = -2\vec{A}$$


Direction
change

(b) Cross product of two vector.

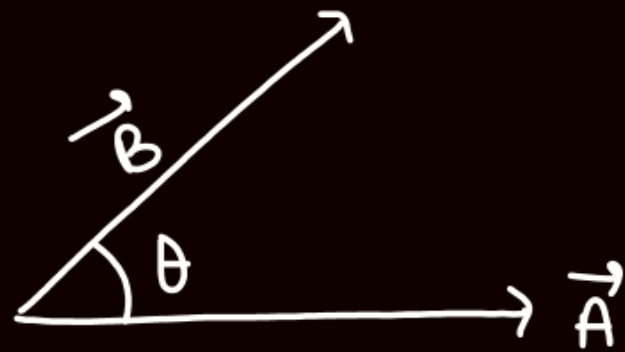
Vector product

3 Mark

2025/2024/2022



The cross product of two vector is the multiplication of two vector and sin of angle b/w them



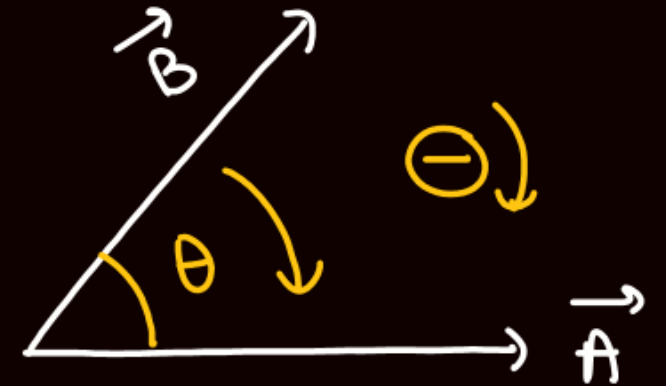
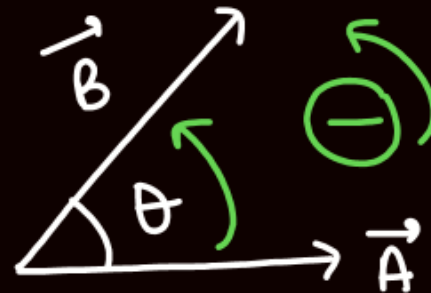
$$\vec{A} \times \vec{B} = \underbrace{AB \sin \theta}_m \underbrace{\hat{n}}_D$$

vector

Example

$$\vec{r} = \vec{\gamma} \times \vec{F}$$
$$\vec{L} = \vec{r} \times \vec{p}$$

① # The Direction of cross product is given by Screw Rule



$$\vec{A} \times \vec{B} = \underline{AB \sin \theta} (\text{outward } \hat{n})$$

$$\vec{B} \times \vec{A} = \underline{BA \sin \theta} (\text{inward})$$

② # $\vec{A} \times \vec{B} \neq \vec{B} \times \vec{A}$ (do not show commutative)

③ # $\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$
(show distributive property)

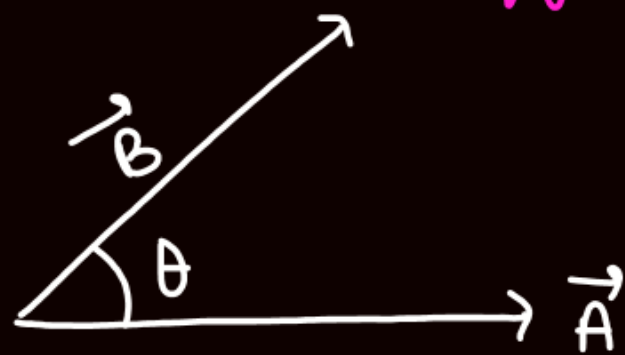
④ # $AB \sin 90^\circ = AB \times 1 = AB$

⑤ # $AB \sin 0 = AB \times 0 = 0$

Rotation

© dot product of two vector.

The dot product of two vector is the multiplication of two vector and $\cos$ of angle b/w them



$$\vec{A} \cdot \vec{B} = \underbrace{AB \cos \theta}_{\text{magnitude}}$$

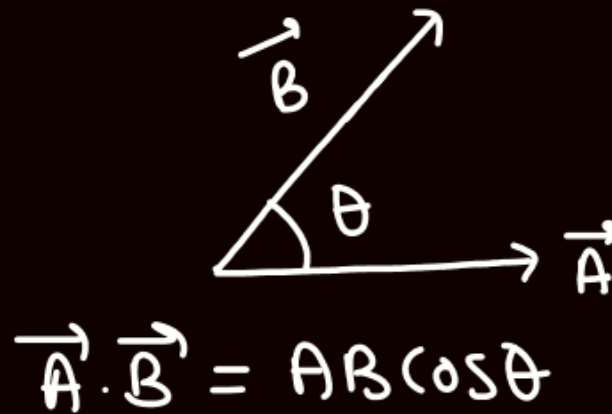
WORK | Energy | Power

$$W = \vec{F} \cdot \vec{x}$$

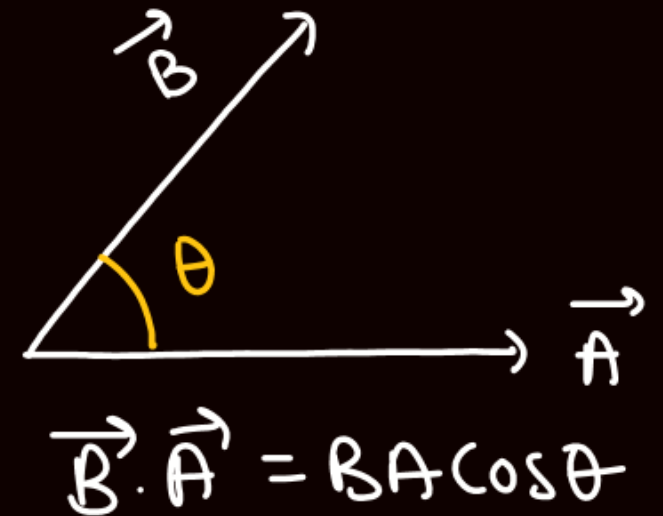
(Scalar product) 3 Mark 2025/2024/2022



① # no need to find the Direction ✓



$$\vec{A} \cdot \vec{B} = AB \cos \theta$$



$$\vec{B} \cdot \vec{A} = BA \cos \theta$$

② # $\vec{A} \times \vec{B} = \vec{B} \times \vec{A}$

(Show commutative)

③ # $\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$

(Show distributive property)

④ # $AB \cos 90^\circ = AB \times 0 = 0$

⑤ # $AB \cos 0 = AB \times 1 = AB$





Multiplication in vectors

3 Mark

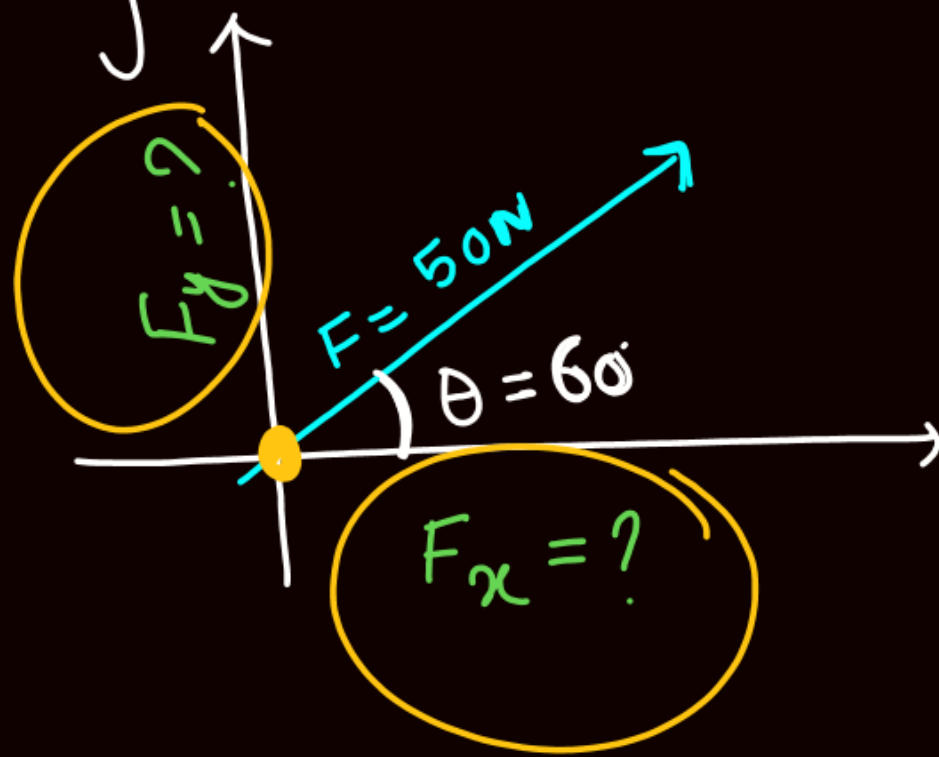
Cross Product ($\vec{A} \times \vec{B}$)

- $\vec{A} \times \vec{B} = AB \sin \theta \hat{n}$
- Vector product
- $\vec{A} \times \vec{B} \neq \vec{B} \times \vec{A}$
- $AB \sin 90^\circ = AB \times 1 = AB$
- $AB \sin 0 = AB \times 0 = 0$

Dot Product ($\vec{A} \cdot \vec{B}$)

- $\vec{A} \cdot \vec{B} = AB \cos \theta$
- Scalar product
- $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$
- $AB \cos 90^\circ = AB \times 0 = 0$
- $AB \cos 0 = AB \times 1 = AB$

Q you have a force of 50N , Find out the Rectangular component of force → 2025



Sol

$$\begin{aligned}
 F_x &= F \cos \theta \\
 &= 50 \cos 60 \\
 &= 50 \times \frac{1}{2} \\
 &= 25 \text{ N } (\hat{i})
 \end{aligned}$$

$$F_y = F \sin \theta$$

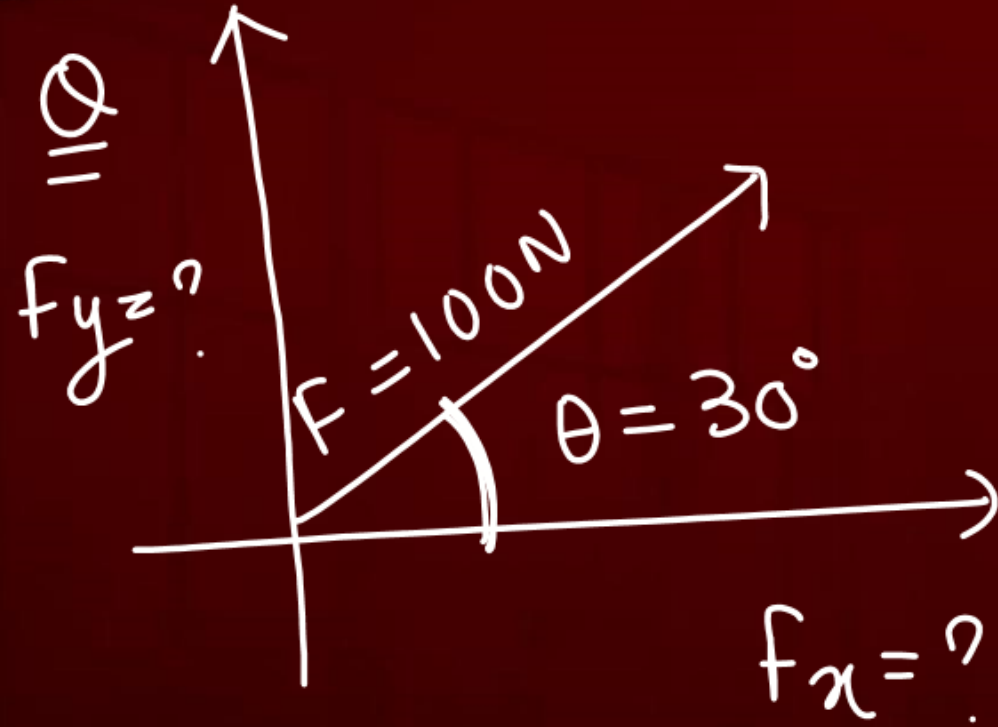
$$F_y = 50 \sin 60$$

$$= 50 \times \frac{\sqrt{3}}{2}$$

$$= 25\sqrt{3} \text{ N } (\hat{j})$$



Homework



**Thank
You**